COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– I Year I Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	BS&H	23SH1T03	Chemistry	3	0	0	3	30	70	100
2	BS&H	23SH1T04	Linear Algebra &Calculus	3	1	0	3	30	70	100
3	ES	23ME1T01	Engineering Graphics	3	0	0	3	30	70	100
4	ES	23EE1T01	Basic Electrical and Electronics Engineering	3	0	0	3	30	70	100
5	ES	23IT1T01	Introduction to Programming	3	0	0	3	30	70	100
6	BS&H	23SH1L03	Chemistry Lab	0	0	2	1		50	50
7	ES	23EE1L01	Electrical and Electronics Engineering Workshop	0	0	3	1.5	30	70	100
8	ES	23IT1L01	Computer Programming Lab	0	1	3	1.5	30	70	100
9	ES	23IT1L02	IT Workshop	-	-	2	1		50	50
10	AC	23PC1P01	Health and Wellness, Yoga and Sports	2	0	0	0.5	-	-	-
	Total Credits						**20.5**	**210**	**590**	**800**

B.Tech.– I Year II Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	BS&H	23SH1T03	Communicative English	2	0	0	2			
2	BS&H	23SH1T04	Engineering Physics	3	0	0	3	30	70	100
3	ES	23ME1T01	Differential equation &Vector Calculus	3	0	0	3	30	70	100
4	ES	23EE1T01	Basic Civil & Mechanical Engineering	3	0	0	3	30	70	100
5	ES	23IT1T01	Data Structures	3	0	0	3	30	70	100
6	BS&H	23SH1L03	Communicative English Lab	0	0	2	1		50	50
7	ES	23EE1L01	Engineering Physics Lab	0	0	2	1		50	50
8	ES	23IT1L01	Data Structures Lab	0	0	3	1.5	30	70	100
9	ES	23IT1L02	Engineering Workshop	0	0	3	1.5	30	70	100
10	AC	23PC1P01	NCC/NSS/ScoutsGuides/Community Service	1	-	-	0.5	-	-	-
	Total Credits						**19.5**	**210**	**590**	**800**

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– II Year I Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	BSC	23SH3T02	Discrete Mathematics & Graph Theory	3	0	0	3	30	70	100
2	PCC	23CS3T01	Software Engineering	3	0	0	3	30	70	100
3	PCC	23CS3T02	Object Oriented Programming Through Java	3	0	0	3	30	70	100
4	PCC	23CS3T03	Advanced Data Structures & Algorithm Analysis	3	0	0	3	30	70	100
5	ESC	23IT3T03	Digital Logic & Computer Organization	3	0	0	3	30	70	100
6	PCC	23CS3L02	Object Oriented Programming Through Java Lab	0	0	3	1.5	30	70	100
7	PCC	23CS3L03	Advanced Data Structures & Algorithm Analysis Lab	0	0	3	1.5	30	70	100
8	SC	23CS3S01	Full Stack Development –I	0	1	2	2	30	70	100
9	MC	23SH3M02	Human Values	2	0	0	-	-	-	-
	Total Credits						**20**	**240**	**560**	**800**

Theory: BSC-1, ESC-1, PC-3, MC-1 **Practical: PC-2, SEC-1**

B.Tech.– II Year II Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	BSC	**23SH4T05**	Managerial Economics and Financial Analysis	2	0	0	2	30	70	100
2	BSC	**23SH4T01**	Probability & Statistics	3	0	0	3	30	70	100
3	PCC	**23IT4T04**	Operating Systems	3	0	0	3	30	70	100
4	PCC	**23CD4T02**	Database Management Systems	3	0	0	3	30	70	100
5	PC	**23CS4T04**	Python Programming	3	0	0	3	30	70	100
6	PCC	**23CD4L02**	Database Management Systems Lab	0	0	3	1.5	30	70	100
7	PCC	**23CS4L04**	Python Programming Lab	0	0	3	1.5	30	70	100
8	SC	**23CS4S02**	Full Stack Development –II	0	1	2	2	-	50	50
9	BSC	**23SH4M01**	Design Thinking & Innovation	0	0	2	1	-	50	50
			Total Credits				**20**	**210**	**590**	**800**
Mandatory Community Service Project Internship of 08 weeks duration during summer vacation										

Theory: BSC-2, PC-3, BS & H-1 **Practical: PC-2, SEC-1**

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COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– III Year I Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	PCC	23CS5T01	Computer Networks	3	0	0	3	30	70	100
2	PCC	**23CS5T02**	Artificial Intelligence	3	0	0	3	30	70	100
3	PCC	23CS5T03	Formal Languages and Automata Theory	3	0	0	3	30	70	100
4	PE-I		1. Object Oriented Analysis and Design	3	0	0	3	30	70	100
		23IT5E03	2. **Data Warehousing & Data Mining**							
			3. Microprocessors & Microcontrollers							
			4. Software Testing Methodologies							
			5. 12-week MOOC Swayam/ NPTEL course recommended by the BoS							
5	OE-I	**23EC5O04**	**Principals of Communication**	3	0	0	3	30	70	100
	OPEN ELECTIVE –I TO OTHER DEPT	23CS5O01	Principles of Operating Systems	3	0	0	3	30	70	100
		23CS5O02	Computer Organization and Architecture	3	0	0	3	30	70	100
6	PCC	23CS5L01	Computer Networks Lab	0	0	3	1.5	30	70	100
7	PCC	23CS5L02	Artificial Intelligence Lab	0	0	3	1.5	30	70	100
8	SEC	23CS5S03	IoT360: From Circuit to Cloud	0	1	2	2	-	100	100
9	ESC	23CS5L03	User Interface Design using Flutter / SWAYAM Plus - Android Application Development (with Flutter)	0	0	2	1	30	70	100
10		23CS5I01	Community Service Project	-	-	-	2	-	100	100
			Total Credits				23	240	760	1000

Theory: PCC-3, PE-1, OE-1 **Practical: PCC-2, SEC-1, ESC-1**

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– III Year II Semester

S.No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	PCC	23CS6T03	Compiler Design	3	0	0	3	30	70	100
2	PCC	23CS6T01	Cryptography & Network Security	3	0	0	3	30	70	100
3	PCC	23IT6T02	Machine Learning	3	0	0	3	30	70	100
4	PE-II	23CS6E01	1. Mobile Adhoc Networks	3	0	0	3	30	70	100
		23CS6E02	2. Cyber Security	3	0	0	3	30	70	100
		23CS6E03	3. DevOps	3	0	0	3	30	70	100
		23CS6E04	4. **Cloud Computing**	3	0	0	3	30	70	100
		23CS6E05	5. 12-week MOOC Swayam/ NPTEL course recommended by the BoS	3	0	0	3	30	70	100
5	PE-III	23CD6E03	1. Software Project Management	3	0	0	3	30	70	100
		23CM6E01	2. Natural Language Processing	3	0	0	3	30	70	100
		23CS6E06	3. **Big Data Analytics**	3	0	0	3	30	70	100
		23IT6E04	4. Distributed Operating System	3	0	0	3	30	70	100
		23CS6E07	512-week MOOC Swayam/ NPTEL course recommended by the BoS	3	0	0	3	30	70	100
6	OE-II	23EC6E09	Introduction to Internet of Things	3	0	0	3	30	70	100
	OPEN ELECTIVE –II TO OTHER DEPT	23CS6O03	Principles of Database Management Systems	3	0	0	3	30	70	100
		23CS6O04	Computer Networks	3	0	0	3	30	70	100
7	PCC	23CS6L01	Cryptography & Network Security Lab	0	0	3	1.5	30	70	100
8	PCC	23IT6L02	Machine Learning Lab	0	0	3	1.5	30	70	100
9	SEC	23SH6S01	Soft skills / / SWAYAM Plus - 21st Century Employability Skills	0	1	2	2	-	100	100
10	ESC	23GE6M01	Technical Paper Writing &IPR	0	0	2	0	-	100	100
	Total Credits						**23**	**240**	**760**	**1000**

Theory: PCC-3, PE-2, OE-1 **Practical: PC-2, SEC-1, ESC-1**

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COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– IV Year I Semester

S. No	Course Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	PCC		Deep Learning	3	0	0	3	30	70	100
2	MS		Human Resources & Project Management	3	0	0	3	30	70	100
3	PE-IV		1. Software Architecture & Design Patterns **2.** Blockchain Technology 3. Augmented Reality & Virtual Reality 4. Quantum Computing 5. 12-week MOOC Swayam/ NPTEL course recommended by the BoS	3	0	0	3	30	70	100
4	PE-V		1. Mobile Computing 2. Agile methodologies 3. Generative AI 4. Computer Vision 5. 12 week MOOC Swayam/ NPTEL course recommended by the BoS	3	0	0	3	30	70	100
5	OE-III			3	0	0	3	30	70	100
6	OE-IV			3	0	0	3	30	70	100
	OPEN ELECTIVE – III TO OTHER DEPT	23CS7O05	Object Oriented Programming Through Java							
	OPEN ELECTIVE – IV TO OTHER DEPT	23CS7O06	Principles of Software Engineering							
7	SEC		Prompt Engineering/ SWAYAM Plus - Certificate program in Prompt Engineering and ChatGPT	0	1	2	2	---	50	50
8	AC		Constitution of India	2	0	0				
9	INTERNSHIP		Evaluation of Industry Internship / Mini Project				2	-	50	50
			Total Credits				**22**	**180**	**520**	**700**
			Mandatory Community Service Project Internship of 08 weeks duration during summer vacation							

Theory: PCC-1, PE-2, OE-2, MS-1 Practical**:** PC-2, SEC-1, ESC-1, AC-1

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE & SYLLABUS – R23

B.Tech.– IV Year II Semester (SEMESTER-VIII)

S No	Category	Course Title	L	T	P	C	IM	EM	TM
THEORY									
1	PCC	Major Project Work, Seminar, Internship in industry	0	0	0	12	60	140	200
Total Credits						**12**	**60**	**140**	**200**

Open Electives, offered to other department students:

Open Elective I: Principles of Operating Systems/ Computer Organization and Architecture
Open Elective II: Principles of Database Management Systems
Open Elective III: Object Oriented Programming Through Java
Open Elective IV: Principles of Software Engineering /Computer Networks

Minor Engineering

Note:

1. To obtain Minor Engineering, student needs to obtain 18 credits by successfully completing any of the following courses in the concern stream.
2. During Minors/Honors Course selection, there should not be any overlapping with Regular/ Major/ OPEN Electives

Minor in CSE

1. Principles of Database Management Systems 3-0-3-4.5 (II-II)
2. Principles of Software Engineering 3-0-0-3 (III-I)
3. Advanced Data Structures & Algorithm Analysis 3-0-3-4.5 (III-II)
4. Principles of Operating Systems 3-0-0-3 (IV-I)
5. Any of the following 12 Week 3 credit NPTEL MOOC Courses
6. Artificial Intelligence: Knowledge Representation and Reasoning
7. Computer Networks and Internet Protocol
8. Machine Learning and Deep Learning - Fundamentals and Applications
9. Fundamentals of Object-Oriented Programming
10. Discrete Mathematics for CS
11. Software Engineering

COURSES OFFERED FOR HONORS DEGREE IN CSE

Note: To obtain Honor's degree, student needs to obtain 18 credits by successfully completing any of the following courses in the concern stream. (without duplication)
Suggested MOOC Courses for HONORS in CSE
Student need to obtain 18 Credits by successfully completing the following
Mandatory Course(s)

1. Parallel Computer Architecture (MOOCS- SWAYAM / NPTEL 12W)
2. Quantum Algorithms and Cryptography (MOOCS- SWAYAM / NPTEL 12W)

Any of the following for remaining 12 Credits

1. Deep Learning for Computer Vision (MOOCS- SWAYAM / NPTEL 12W)
2. Applied Linear Algebra in AI & ML (MOOCS- SWAYAM / NPTEL 12W)
3. Applied Time-Series Analysis (MOOCS- SWAYAM / NPTEL 12W)
4. Machine Learning for Engineering and Science Applications (MOOCS- SWAYAM / NPTEL 12W)
5. Practical High-Performance Computing (MOOCS- SWAYAM / NPTEL 12W)
6. Deep Learning for Natural Language Processing (MOOCS- SWAYAM / NPTEL 12W)
7. Privacy and Security in Online Social Media (MOOCS- SWAYAM / NPTEL 12W)
8. Natural Language Processing (MOOCS- SWAYAM / NPTEL 12W)

III Year I Semester	Computer Networks	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the Course, students will be able to:
CO1: Outline Network models and Transmission media.
CO2: Compare various error control and flow control concepts.
CO3: Analyze MAC layer protocols and LAN technologies
CO4: Summarize various Routing algorithms and Congestion control principles.
CO5: Describe Transport layer protocols and Application layer protocols

UNIT I: Introduction: Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP.
Physical Layer –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and introduction about Unguided Media.

UNIT II: Data link layer: Design issues, **Framing**: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, **Checksum:** idea, one's complement internet checksum, services provided to Network Layer, **Elementary Data Link Layer protocols**: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.
Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC, Point to Point Protocol (PPP)

UNIT III: Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, **Controlled Access:** Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA).
Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet (100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.

UNIT IV: The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks,
Routing Algorithms-The Optimality Principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, **Congestion Control Algorithms**-General principles of congestion control, Congestion prevention polices, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm-Leaky bucket & Token bucket.
Internet Working: How networks differ- How networks can be connected- Tunnelling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6.

UNIT V: The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol-User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features- Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP.
Application Layer –- World Wide Web: HTTP, Electronic Mail, TELENET, Domain Name System.

Text Books:

1. Computer Networks, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH.

References Books:

1. Data Communications and Networks- Achut S Godbole, AtulKahate
2. Computer Networks, Mayank Dave, CENGAGE

E-Resources:

https://onlinecourses.nptel.ac.in/noc22_cs19/preview

III Year I Semester	Artificial Intelligence	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the Course, students will be able to

CO1: Describe the historical development, motivations, and milestones in Artificial Intelligence.

CO2: Apply heuristic-based search algorithms to solve problems with partial information and demonstrate the use of game-playing algorithms.

CO3: Define and describe various approaches to knowledge representation

CO4: Apply inference techniques in First-Order Logic to derive logical conclusions and implement the basic reinforcement learning algorithms for decision-making

CO5: Illustrate the working of classical expert systems like MYCIN, DART, and XCON, and the use of Expert System Shells.

UNIT I: Introduction: History, intelligent systems, foundations of AI, applications, tic-tac-toe game playing, development of AI languages, current trends.

UNIT II: Searching: Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* , AO* Algorithms.
Problem reduction and game playing: Problem reduction, Game Playing- Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning.

UNIT III: Representation of Knowledge: Knowledge representation issues, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempster shafer theory.

UNIT IV: Logic concepts: First order logic. Inference in first order logic, propositional vs first order inference, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

UNIT V: Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Text Books:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", Second Edition, Pearson Education.
2. Artificial Intelligence- Saroj Kaushik, CENGAGE Learning
3. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, "Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problem solving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

E-Resources:

1. https://ai.google/
2. https://swayam.gov.in/nd1_noc19_me71/preview

III Year I Semester	Formal Languages and Automata Theory	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the course, students will be able to
CO1: Understand the basic concepts of formal languages of finite automata techniques.
CO2: Solve regular expressions and various problems to minimize FA.
CO3: Apply various languages to construct context free grammar.
CO4: Apply Normal forms and Push down automata to solve various language acceptance problems.
CO5: Apply Turing Machine to solve various problems related to calculations.

Unit I: Finite Automata
Need of Automata theory, Central Concepts of Automata Theory, Automation, Finite Automation, Transition Systems, Acceptance of a String, DFA, Design of DFAs, NFA, Design of NFA, Equivalence of DFA and NFA, Conversion of NFA into DFA, Finite Automata with Є-Transitions, Minimization of Finite Automata, Finite Automata with output-Mealy and Moore Machines, Applications and Limitation of Finite Automata.

Unit II: Regular Expressions
Regular Expressions, Regular Sets, Identity Rules, Equivalence of two RE, Manipulations of REs, Finite Automata and Regular Expressions, Inter Conversion, Equivalence between FA and RE, Pumping Lemma of Regular Sets, Closure Properties of Regular Sets, Grammars, Classification of Grammars, Chomsky Hierarchy Theorem, Right and Left Linear Regular Grammars, Equivalence between RG and FA, Inter Conversion.

Unit III: Context Free Grammar
Formal Languages, Context Free Grammar, Leftmost and Rightmost Derivations, Parse Trees, Ambiguous Grammars, Simplification of Context Free Grammars-Elimination of Useless Symbols, Є-Productions and Unit Productions, Normal Forms-Chomsky Normal Form and Greibach Normal Form, Pumping Lemma, Closure Properties, Applications of Context Free Grammars.

Unit IV: Pushdown Automata
Pushdown Automata, Definition, Model, Graphical Notation, Instantaneous Description, Language Acceptance of Pushdown Automata, Design of Pushdown Automata, Deterministic and Non – Deterministic Pushdown Automata, Equivalence of Pushdown Automata and Context Free Grammars, Conversion, Two Stack Pushdown Automata, Application of Pushdown Automata.

Unit V: Turning Machine
Definition, Model, Representation of TMs-Instantaneous Descriptions, Transition Tables and Transition Diagrams, Language of a TM, Design of TMs, Types of TMs, Church's Thesis, Universal and Restricted TM, Decidable and Un-decidable Problems, Halting Problem of TMs, Post's Correspondence Problem, Modified PCP, Classes of P and NP, NP-Hard and NP-Complete Problems.

Text Books:
1. Introduction to Automata Theory, Languages and Computation, J. E. Hopcroft, R. Motwani and J. D. Ullman, 3rd Edition, Pearson, 2008.
2. Theory of Computer Science-Automata, Languages and Computation, K. L. P. Mishra and N. Chandrasekharan, 3rd Edition, PHI, 2007.

Reference Books:

1. Elements of Theory of Computation, Lewis H.P. & Papadimitriou C.H., Pearson /PHI.
2. Theory of Computation, V. Kulkarni, Oxford University Press, 2013.
3. Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill, 2014.

E-Resources:

1. https://nptel.ac.in/courses/106/104/106104028/
2. https://www.youtube.com/playlist?list=PL85CF9F4A047C7BF7

III Year I Semester	(Professional Elective-I) 1. Object Oriented Analysis And Design	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the course, students will be able to:

CO1: Find solutions to the complex problems using object-oriented approach

CO2: Represent classes, responsibilities and states using UML notation

CO3: Identify classes and responsibilities of the problem domain

CO4: Develop and explore the behavioral model

CO5: Apply the concepts of behavioral and architectural models to create comprehensive design representations for software systems.

UNIT I: Introduction: The Structure of Complex systems, The Inherent Complexity of Software, Attributes of Complex System, Organized and Disorganized Complexity, Bringing Order to Chaos, Designing Complex Systems. **Case Study:** System Architecture: Satellite-Based Navigation

UNIT II: Introduction to UML: Importance of modeling, principles of modeling, object oriented modeling, conceptual model of the UML. **Basic Structural Modeling:** Classes, Relationships, common Mechanisms, and diagrams. **Case Study:** Control System: Traffic Management.

UNIT III: Class & Object Diagrams: Terms, concepts, modeling techniques for Class & Object Diagrams. **Advanced Structural Modeling:** Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. **Case Study:** AI: Cryptanalysis.

UNIT IV: Basic Behavioral Modeling-I: Interactions, Interaction diagrams Use cases, Use case Diagrams, Activity Diagrams. **Case Study:** Web Application: Vacation Tracking System

UNIT V: Advanced Behavioral Modeling: Events and signals, state machines, processes and Threads, time and space, state chart diagrams.
Architectural Modeling: Component, Deployment, Component diagrams and Deployment diagrams.
Case Study: Weather Forecasting

Text Books:

1. Grady BOOCH, Robert A. Maksimchuk, Michael W. ENGLE, Bobbi J. Young, Jim Conallen, Kellia Houston , "Object- Oriented Analysis and Design with Applications", 3rd edition, 2013, PEARSON.
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Pearson Education.

Reference Books:

1. Meilir Page-Jones: Fundamentals of Object Oriented Design in UML, Pearson Education.
2. Pascal Roques: Modeling Software Systems Using UML2, WILEY- Dreamtech India Pvt. Ltd.
3. Atul Kahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
4. Appling UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, Craig Larman, Pearson Education.

E-Resources:

1. https://archive.nptel.ac.in/courses/106/105/106105153/

III Year I Semester	(Professional Elective-I) 2.Data Warehousing & Data Mining	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the course, students will be able to

CO1: Understand the basic concepts of data warehousing, OLAP, data cube modelling, and cloud-based data warehouse architectures.

CO2: Apply techniques for data cleaning, integration, transformation, and reduction to prepare high-quality data for mining.

CO3: Analyze and implement classification techniques including decision trees, Bayesian classifiers, and rule-based models for supervised learning.

CO4: Demonstrate the ability to perform association analysis using algorithms like Apriori and FP-Growth and evaluate rule interestingness.

CO5: Apply clustering methods such as K-means, Hierarchical, and DBSCAN to discover patterns and groupings in datasets.

UNIT I: Data Warehousing and Online Analytical Processing: Basic concepts, Data Warehouse Modeling: Data Cube and OLAP, Data Warehouse Design and Usage, Data Warehouse Implementation, Data Mining and Patten Mining, Technologies, Applications, Major issues, Data Objects & Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity.

UNIT II: Data Preprocessing: An Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization.

UNIT III: Classification: Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction, Visual Mining for Decision Tree Induction, Bayesian Classification Methods: Bayes Theorem, Naïve Bayes Classification, Rule-Based Classification, Model Evaluation and Selection.

UNIT IV: Association Analysis: Problem Definition, Frequent Itemset Generation, Rule Generation: Confident Based Pruning, Rule Generation in Apriori Algorithm, Compact Representation of frequent item sets, FP-Growth Algorithm.

UNIT V: Cluster Analysis: Overview, Basics and Importance of Cluster Analysis, clustering techniques, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bi-secting K Means, Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses.

Text Books:

1. Data Mining concepts and Techniques, 3rd edition, Jiawei Han, Michel Kamber, Elsevier, 2011.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson, 2012.

Reference Books:

1. Data Mining: VikramPudi and P. Radha Krishna, Oxford Publisher.
2. Data Mining Techniques, Arun K Pujari, 3rd edition, Universities Press,2013.

E-Resources:

1. (NPTEL course by Prof.PabitraMitra) http://onlinecourses.nptel.ac.in/noc17_mg24/preview
2. http://www.saedsayad.com/data_mining_map.htm
3. https://www.saedsayad.com/data_mining.htm
4. https://www.tutorialspoint.com/data_mining/index.htm

III Year I Semester	(Professional Elective-I) 3. Microprocessors & Microcontrollers	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the course, students will be able to

CO1: Understand and explain the architecture of the 8086 microprocessor, including its pin configuration, internal blocks, and system operation modes.

CO2: Develop assembly language programs for the 8086 using appropriate instructions, addressing modes, and assembler directives.

CO3: Interface various peripheral devices with the 8086 microprocessor and implement interrupt-driven and DMA-based data transfer systems.

CO4: Explain the architecture and instruction set of the 8051 microcontroller and develop basic assembly language programs using different addressing modes.

CO5: Implement peripheral interfacing with the 8051 microcontroller for real-time applications and compare different processor architectures including PIC and ARM.

UNIT I - 8086 Architecture: Main features, pin diagram/description, 8086 microprocessor family, internal architecture, bus interfacing unit, execution unit, interrupts and interrupt response, 8086 system timing, minimum mode and maximum mode configuration.

UNIT II - 8086 Programming: Program development steps, instructions, addressing modes, assembler directives, writing simple programs with an assembler, assembly language program development tools.

UNIT III - 8086 Interfacing: Semiconductor memories interfacing (RAM, ROM), Intel 8255 programmable peripheral interface, Interfacing switches and LEDS, Interfacing seven segment displays, software and hardware interrupt applications, Intel 8251 USART architecture and interfacing, Intel 8237a DMA controller, stepper motor, A/D and D/A converters, Need for 8259 programmable interrupt controllers.

UNIT IV: Microcontroller, Architecture of 8051, Special Function Registers(SFRs), I/O Pins Ports and Circuits, Instruction set, Addressing modes, Assembly language programming.

UNIT V: Interfacing Microcontroller, Programming 8051 Timers, Serial Port Programming, Interrupts Programming, LCD & Keyboard Interfacing, ADC, DAC & Sensor Interfacing, External Memory Interface, Stepper Motor and Waveform generation, Comparison of Microprocessor, Microcontroller, PIC and ARM processors

Textbooks:

1. Microprocessors and Interfacing – Programming and Hardware by Douglas V Hall, SSSP Rao, Tata McGraw Hill Education Private Limited, 3rd Edition,1994.
2. K M Bhurchandi, A K Ray, Advanced Microprocessors and Peripherals, 3rd edition, McGraw Hill Education, 2017.
3. Raj Kamal, Microcontrollers: Architecture, Programming, Interfacing and System Design, 2nd edition, Pearson, 2012.

Reference Books:

1. Ramesh S Gaonkar, Microprocessor Architecture Programming and Applications with the 8085, 6th edition, Penram International Publishing, 2013.
2. Kenneth J. Ayala, The 8051 Microcontroller, 3rd edition, Cengage Learning, 2004.

E-Resources:

1. https://archive.nptel.ac.in/courses/108/105/108105102/

III Year II Semester	(Professional Elective-I) Software Testing Methodologies	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the Course, students will be able to

CO1: Understand the purpose of software testing, the nature of software bugs, and apply path testing techniques using flow graphs and predicates.

CO2: Apply transaction flow, data flow, and domain testing strategies to evaluate the functional correctness and testability of software applications.

CO3: Analyze and implement test cases using path products, regular expressions, and logic-based testing techniques such as decision tables and KV charts.

CO4: Evaluate and design tests based on state graphs and transition testing techniques to identify state-related defects and improve testability.

CO5: Construct and interpret graph matrices and apply node reduction algorithms while gaining hands-on experience with testing tools like JMeter, Selenium, SoapUI, or Catalon.

UNIT I

Introduction: Purpose of testing, Productivity and Quality in Testing, Goals of Testing Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.

UNIT II

Transaction Flow Testing: transaction flows, transaction flow testing techniques. Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing. Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.

UNIT III

Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: overview, decision tables, path expressions, KV charts, specifications.

UNIT IV

State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips.

UNIT V

Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like JMeter/selenium/SoapUI/Catalon).

Text Books:

1. Software Testing techniques - Baris Beizer, Dreamtech, second edition.
2. Software Testing Tools – Dr. K. V. K. K. Prasad, Dreamtech.

Reference Books:

1. The craft of software testing - Brian Marick, Pearson Education.
2. Software Testing Techniques – SPD(Oreille)

3. Software Testing in the Real World – Edward Kit, Pearson.
4. Effective methods of Software Testing, Perry, John Wiley.
5. Art of Software Testing – Meyers, John Wiley.

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc22_cs61/preview
2. https://nptel.ac.in/courses/106105087
3. https://www.coursera.org/learn/logic-introduction
4. https://nptel.ac.in/courses/111106050
5. https://www.coursera.org/learn/software-testing
6. https://www.coursera.org/specializations/software-design-architecture

III Year I Semester	Computer Networks Lab	L	T	P	C
		0	0	3	1.5

Course Outcomes:

At the end of the course, students will be able to:

CO1: Apply the basics of Physical layer in real time applications
CO2: Apply data link layer concepts, design issues, and protocols
CO3: Apply Network layer routing protocols and IP addressing
CO4: Apply the Transport Layer paradigms and protocols.
CO5: Simulate and demonstrate the performance using NS2.

List of Experiments:

1. Study of Network devices in detail and connect the computers in Local Area Network.
2. Write a Program to implement the data link layer farming methods such as
   ii) Character stuffing ii) bit stuffing.
3. Write a Program to implement data link layer farming method checksum.
4. Write a program for Hamming Code generation for error detection and correction.
5. Write a Program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
6. Write a Program to implement Sliding window protocol for Go back N.
7. Write a Program to implement Sliding window protocol for Selective repeat.
8. Write a Program to implement Stop and Wait Protocol.
   a. Write a program for congestion control using leaky bucket algorithm
9. Write a Program to implement Dijkstra 's algorithm to compute the Shortest path through a graph.
10. Write a Program to implement Distance vector routing algorithm by obtaining routing table at each node (Take an example subnet graph with weights indicating delay between nodes).
11. Write a client-server application for chat using UDP
12. Write a Program to implement Broadcast tree by taking subnet of hosts.
13. Do the following using NS2 Simulator
    i. NS2 Simulator-Introduction
    ii. Simulate to Find the Number of Packets Dropped
    iii. Simulate to Find the Number of Packets Dropped by TCP/UDP
    iv. Simulate to Find the Number of Packets Dropped due to Congestion

Text books:

1. Kurose, J. F., & Ross, K. W. (2020). *Computer networking: A top-down approach* (7th ed.). Pearson.
2. Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer networks* (5th ed.). Pearson Education.
3. Stallings, W. (2021). *Data and computer communications* (11th ed.). Pearson.
4. Donahoo, M. J., & Calvert, K. L. (2009). *TCP/IP sockets in C: Practical guide for programmers* (2nd ed.). Morgan Kaufmann.

Reference books:

1. Aboelela, E. (2003). *Network simulation experiments manual* (2nd ed.). Morgan Kaufmann.
2. Forouzan, B. A. (2017). *Data communications and networking* (5th ed.). McGraw-Hill Education.

E-Resources:

1.https://www.isi.edu/nsnam/ns/

III Year I Semester	Artificial Intelligence Lab	L	T	P	C
		0	0	3	1.5

Course Outcomes:
At the end of the course, students will be able to
CO1: Study the concepts of Artificial Intelligence
CO2: Learn the methods of solving problems using Artificial Intelligence
CO3: Implement Informed and uninformed Search Strategies.
CO4: Implement gaming and puzzle applications
CO5: Design and implement expert system application Applying

LIST OF PROGRAMS:
1. Study of Prolog.
2. Write a program to implement factorial, Fibonacci of a given number and Towers of Hanoi Problem using LISP/PROLOG
3. Write a python program to implement a Water Jug Problem?
4. Write a program to Implement Breadth First Search using Python.
5. Write a program to Implement Depth First Search using Python.
6. Write a python program to implement A*algorithm.(Ex:find the shortest path)
7. Write a program to solve traveling salesman problem.
8. Write a python program to implement the alpha-beta pruning(Ex:Tic-Tac-Toe game).
9. Implement the Constraint Specific Problem(Ex: crossword puzzle).
10. Design of Intelligent systems.
11. Write a program to implement odd and even magic squares of nxn dimension.
12. Write a python program to implement then-queens problem.
13. Solve 8-puzzle problem using Best First Search.
14. Implement a basic Dempster-Shafer theory example by combining evidence from multiple sources to determine the likelihood of an event and use two pieces of evidence to update the belief in a hypothesis
15. Write a case study on MYCIN, DART, or XCON

Text Books:
1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", SecondEdition, Pearson Education.
2. Artificial Intelligence- Saroj Kaushik, CENGAGE Learning
3. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill

Reference Books:
1. David Poole, Alan Mackworth, Randy Goebel,"Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problemsolving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, SarojKaushik, CENGAGE Learning.

E-Resources:
1. https://www.studocu.com/in/document/dr-apj-abdul-kalam-technical-university/artificial-intelligence-lab/artificial-intelligence-lab-files/16507070
2. https://www.tutorialspoint.com/prolog/prolog_towers_of_hanoi_problem.htm

KKR & KSR INSTITUUTE OF TECHNOLOGY AND SCIENCES (AUTONOMOUS)
(Accredited by NBA & NAAC with Grade "A" and Affiliated to JNTUK-Kakinada)
(Vinjanampadu, Vatticherukuru Mandal, Guntur-522017.Andhra Pradesh. INDIA)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III Year I Semester	SEC– IoT360: From Circuit to Cloud	L	T	P	C
		0	1	2	2

Course Outcomes:
At the end of the course, students will be able to

CO1: Demonstrate understanding of IoT concepts, architecture, and the role of microcontrollers such as ESP32/Node MCU in IoT systems.
CO2: Implement basic input/output interfacing techniques using sensors and actuators to collect and respond to environmental data.
CO3: Connect and configure IoT devices to cloud platforms like ThingSpeak and Firebase for real-time data exchange and visualization.
CO4: Develop mobile-controlled IoT systems using Blynk for remote monitoring and device control via smartphones.
CO5: Design and demonstrate a complete IoT-based automation project integrating sensors, actuators, cloud services, and communication apps.

Experiments covering the Topics:

- Hardware, Sensors, Microcontroller Programming
- Cloud Integration, Data Monitoring, Automation

List of Experiments

Experiment 1: IoT Introduction & Hardware Setup

a) Blink an LED - Digital output basics.
b) Button Controlled LED - Reading digital input.
c) Read Potentiometer Value - Analog input with serial output.
d) Light Detection with LDR - Automate LED based on light level.

Experiment 2: Sensors and Actuators

a) DHT11: Temperature & Humidity Monitor
b) PIR Motion Detection with Buzzer Alert
c) Soil Moisture Monitoring
d) Servo Motor Control with Potentiometer
e) Control Appliance using Relay
f) Display Sensor Data on OLED

Experiment 3: Wi-Fi & ThingSpeak Cloud Integration

a) Send DHT11 Data to ThingSpeak
b) Visualize Soil Moisture Data on Cloud

Experiment 4: Firebase & Real-Time Cloud Communication

a) Push Gas Sensor Data to Firebase
b) Read Firebase Values and Display on OLED

Experiment 5: Remote Monitoring & Control with Blynk

a) Control Relay from Blynk App
b) Live Temp/Humidity on Blynk Interface

Experiment 6: Automation and Capstone Project

a) Motion Alert via Email using IFTTT
b) Control LED via Telegram Bot
c) Smart Plant Monitoring System
d) Weather Dashboard with Cloud Logging

Text Books:

1. Norris, D. (2016). The Internet of Things: Do-it-yourself at home projects for Arduino, Raspberry Pi, and BeagleBone Black. McGraw-Hill Education.
2. Schwartz, M. (2014). Internet of Things with the Arduino Yun. Packt Publishing.
3. Kumar, R., & Raj, P. (2020). The Internet of Things: Architecture and applications. Wiley India.

Reference Books:

1. Raj, P., & Raman, A. C. (2017). *The Internet of Things: Enabling technologies, platforms, and use cases*. CRC Press.
2. Pillai, A. S. (2019). *Internet of Things using Arduino and Blynk*. Notion Press.

E-Resources:

- https://www.arduino.cc/en/Guide
- https://thingspeak.com/docs

III Year I Semester	User Interface Design using Flutter	L	T	P	C
		0	0	2	1

Course Outcomes:

At the end of the course, students will be able to:

CO1: Implement Flutter Widgets and Layouts.
CO2: Design Responsive UI with Navigation in Flutter.
CO3: Create custom widgets for specific UI elements and apply styling using themes and custom styles.
CO4: Design a form with various input fields, along with validation and error handling.
CO5: Fetches data and write code for unit Test for UI components and also animation.

List of Experiments

1. a) Install Flutter and Dart SDK.
   b) Write a simple Dart program to understand the language basics.
2. a) Explore various Flutter widgets (Text, Image, Container, etc.).
   b) Implement different layout structures using Row, Column, and Stack widgets.
3. a) Design a responsive UI that adapts to different screen sizes.
   b) Implement media queries and breakpoints for responsiveness.
4. a) Set up navigation between different screens using Navigator.
   b) Implement navigation with named routes.
5. a) Learn about stateful and stateless widgets.
   b) Implement state management using set State and Provider.
6. a) Create custom widgets for specific UI elements.
   b) Apply styling using themes and custom styles.
7. a) Design a form with various input fields.
   b) Implement form validation and error handling.
8. a) Add animations to UI elements using Flutter's animation framework.
   b) Experiment with different types of animations (fade, slide, etc.).
9. a) Fetch data from a REST API.
   b) Display the fetched data in a meaningful way in the UI.
10. a) Write unit tests for UI components.
    b) Use Flutter's debugging tools to identify and fix issues.

Text Books:

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.
2. Rap Payne, Beginning App Development with Flutter: Create Cross-Platform Mobile Apps 1st Edition, Apres.
3. Richard Rose, Flutter & Dart Cookbook, Developing Full stack Applications for the Cloud, Oreilly.

Reference Books:

1. Learning Continuous Integration with Jenkins: A beginner's guide to implementing Continuous Integration and Continuous Delivery using Jenkins - Nikhil Pathania ,Packt publication.
2. Jenkins 2 – Up and Running: Evolve Your Deployment Pipeline for Next Generation Automation - Brent Laster, O'Reilly publication.

E-Resources:

1. infyspringboard.onwingspan.com/web/en/page/lex_auth_01366558964890009633
2. infyspringboard.onwingspan.com/en/viewer/html/lex_auth_013538989171924992269_shared

III Year II Semester	Compiler Design	L	T	P	C
		3	0	0	3

Course Code: 23CS6T03

Course Outcomes:
At the end of the course, Students will be able to
CO1: Understand the structure of a compiler and design lexical analyzers using finite automata and LEX.
CO2: Analyze and implement top-down and bottom-up parsing techniques with error handling.
CO3: Apply syntax-directed translation to generate intermediate code and perform semantic analysis.
CO4: Understand and apply code optimization techniques to enhance program performance.
CO5: Understand run-time environments and generate efficient target code with register allocation.

UNIT I: Lexical Analysis: Language Processors, Structure of a Compiler, Lexical Analysis, The Role of the Lexical Analyzer, Bootstrapping, Input Buffering, Specification of Tokens, Recognition of Tokens, Lexical Analyzer Generator-LEX, Finite Automata, Regular Expressions and Finite Automata, Design of a Lexical Analyzer Generator.
Syntax Analysis: The Role of the Parser, Context-Free Grammars, Derivations, Parse Trees, Ambiguity, Left Recursion, Left Factoring,

UNIT II: Top Down Parsing: Pre-Processing Steps of Top Down Parsing, Backtracking, Recursive Descent Parsing, LL (1) Grammars, Non-recursive Predictive Parsing, Error Recovery in Predictive Parsing.
Bottom Up Parsing: Introduction, Difference between LR and LL Parsers, Types of LR Parsers, Shift Reduce Parsing, SLR Parsers, Construction of SLR Parsing Tables, More Powerful LR Parses, Construction of CLR (1) and LALR Parsing Tables, Dangling Else Ambiguity, Error Recovery in LR Parsing, Handling Ambiguity Grammar with LR Parsers.

UNIT III: Syntax Directed Translation: Syntax-Directed Definitions, Evaluation Orders for SDD's, Applications of Syntax Directed Translation, Syntax-Directed Translation Schemes, Implementing L-Attributed SDD's. **Intermediate Code Generation:** Variants of Syntax Trees, Three Address Code, Types and Declarations, Translation of Expressions, Type Checking, Control Flow, Backpatching, Intermediate Code for Procedures.

UNIT IV: Code Optimization: The Principle Sources of Optimization, Basic Blocks, Optimization of Basic Blocks, Structure Preserving Transformations, Flow Graphs, Loop Optimization, Data-Flow Analysis, Peephole Optimization

UNIT V: Run Time Environments: Storage Organization, Run Time Storage Allocation, Activation Records, Procedure Calls, Displays **Code Generation:** Issues in the Design of a Code Generator, Object Code Forms, Code Generation Algorithm, Register Allocation and Assignment.

Text Books:
1. Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman, Pearson, 2007.
Reference Books:
1. Compiler Construction, Principles and Practice, Kenneth C Louden, Cengage Learning, 2006
2. Modern compiler implementation in C, Andrew W Appel, Revised edition, Cambridge University Press.

3. Optimizing Compilers for Modern Architectures, Randy Allen, Ken Kennedy,Morgan Kauffmann, 2001.
4. Levine, J.R., T. Mason and D. Brown, Lex and Yacc, edition, O'Reilly & Associates, 1990

E-Resources:

https://onlinecourses.nptel.ac.in/noc21_cs07/preview

III Year II Semester	Cryptography & Network Security	L	T	P	C
		3	0	0	3

Course Code: 23CS6T01
Course Outcomes:
At the end of the course, students will be able to:
CO1: Understand classical encryption techniques and identify various cyber threats and vulnerabilities and apply number theory concepts to support and analyze cryptographic algorithms.
CO2: Analyze and implement symmetric and public key cryptographic algorithms for secure communication.
CO3: Evaluate cryptographic hash functions and digital signature schemes for data integrity and authentication.
CO4: Understand authentication mechanisms, email security protocols, and IP security architectures.
CO5: Apply network security basics, analyze different attacks on networks and evaluate the performance of firewalls and security protocols like SSL, IPSec, and PGP

UNIT I: Classical Encryption Techniques: Security Attacks, Services & Mechanisms, Symmetric Cipher Model. Cyber Threats, Phishing Attack, Web Based Attacks, SQL Injection Attacks, Buffer Overflow& Format String Vulnerabilities, TCP session hijacking, UDP Session Hijacking.
Number Theory: Prime and Relatively Prime Numbers, Modular Arithmetic, Fermat's and Euler's Theorems, the Chinese Remainder Theorem, Discrete Logarithms.

UNIT II: Block Ciphers: Traditional Block Cipher Structure, Block Cipher Design Principles.
Symmetric Key Cryptography: Data Encryption Standard (DES), Advanced Encryption Standard (AES), Blowfish, IDEA, Block Cipher Modes of Operations.
Public Key Cryptography: Principles, Public Key Cryptography Algorithms, RSA Algorithm, Diffie-Hellman Key Exchange, Elliptic Curve Cryptography.

UNIT III: Cryptographic Hash Functions: Application of Cryptographic Hash Functions, Requirements & Security, Secure Hash Algorithm, Message Authentication Functions, Requirements & Security, HMAC & CMAC.
Digital Signatures: NIST Digital Signature Algorithm, Key Management and Distribution.

UNIT IV: User Authentication: Remote User Authentication Principles, Kerberos. Electronic Mail Security: Pretty Good Privacy (PGP) And S/MIME.
IP Security: IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management.

UNIT V: Transport Level Security: Web Security Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Shell (SSH)
Firewalls: Characteristics, Types of Firewalls, Placement of Firewalls, Firewall Configuration, Trusted Systems.

Text Books:

1. Cryptography and Network Security- William Stallings, Pearson Education, 7th Edition.
2. Cryptography, Network Security and Cyber Laws – Bernard Menezes, Cengage Learning, 2010 edition.

Reference Books:

1. Cryptography and Network Security-BehrouzAForouzan, ,Debdeep Mukhopadhyaya, Mc-GrawHill, 3rd Edition, 2015.
2. Network Security Illustrated,Jason Albanese andWes Sonnenreich, MGH Publishers, 2003.

E-Resources:

1. https://nptel.ac.in/courses/106/105/106105031/lecture by Dr.Debdeep Mukhopadhyay IIT Kharagpur [Video Lecture]
2. https://nptel.ac.in/courses/106/105/106105162/lecture by Dr. Sourav Mukhopadhyay IIT Kharagpur [Video Lecture]
3. https://www.mitel.com/articles/web-communication-cryptography-and-network-security[web articles by Mitel Power Connections].
4. NPTEL :https://nptel.ac.in/courses/106105031/lecture by Dr.Debdeep Mukhopadhyay IIT Kharagpur [Video Lecture]
5. Text Book free access:: https://williamstallings.com/Crypto/Crypto6e.html

III Year II Semester	Machine Learning	L	T	P	C
		3	0	0	3

Course Code: 23IT6T02

Course Outcomes:

At the end of the course, students will be able to:

CO1: Identify machine learning techniques suitable for a given problem

CO2: Implement classification algorithms based on distance measures such as KNN and Radius Distance

CO3: Develop solutions for decision problems using Bayesian and instance-based learning.

CO4: Identify and handle linearly non-separable cases using techniques such as Non-linear SVMs and the Kernel Trick.

CO5: Analyze and implement clustering algorithms including K-Means, Fuzzy C-Means, Rough K-Means, and Expectation Maximization, to discover patterns in unlabeled data.

UNIT I:

Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by Rote, Learning by Induction, Reinforcement Learning, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets.

UNIT II:

Nearest Neighbour-Based Models: Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Different Classification Algorithms Based on the Distance Measures, K-Nearest Neighbour Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

UNIT III:

Models Based on Decision Trees: Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Random Forests for Classification and Regression.

The Bayes Classifier: Introduction to the Bayes Classifier, Bayes' Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification, Class Conditional Independence and Naive Bayes Classifier (NBC)

UNIT IV:

Linear Discriminants for Machine Learning: Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs).

UNIT V:

Clustering: Introduction to Clustering, Partitioning of Data, Matrix Factorization, Clustering of Patterns, Divisive Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Spectral Clustering.

Text Books:

1. "Machine Learning Theory and Practice", M N Murthy, V S Ananth Narayana, Universities Press (India), 2024

Reference Books:

1. "Machine Learning", Tom M. Mitchell, McGraw-Hill Publication, 2017
2. "Machine Learning in Action",Peter Harrington, DreamTech
3. Applied Machine Learning, 2nd edition, 2021, M. Gopal
4. "Introduction to Data Mining", Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7thEdition, 2019.

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc23_cs18/preview
2. Andrew Ng, "Machine Learning Yearning" https://www.deeplearning.ai/machine-learning-yearning/
3. ShaiShalev-Shwartz ,Shai Ben-David, "Understanding Machine Learning: From Theory to Algorithms", Cambridge University Press
4. https://www.cse.huji.ac.il/~shais/UnderstandingMachineLearning/index.html

III Year II Semester	1.Mobile Adhoc Networks (Professional Elective-II)	L	T	P	C
		3	0	0	3

Course Outcomes:

At the end of the course, students will be able to

CO1: Explain the characteristics, applications, and MAC protocol design issues of Mobile Ad Hoc Networks (MANETs).

CO2: Analyze and compare routing and transport layer protocols used in ad hoc wireless networks to ensure efficient data communication.

CO3: Analyze the security challenges in MANETs and design secure routing and intrusion detection mechanisms to protect the network.

CO4: Apply the concepts of wireless sensor network architecture, including sensor design, communication, and data retrieval techniques for practical applications.

CO5: Understand and implement security protocols, key management, and simulation tools for secure operation of wireless sensor networks**.**

UNIT I:

Introduction to Ad Hoc Wireless Networks- Cellular and Ad Hoc Wireless Networks, Characteristics of MANETs, Applications of MANETs, Issues and Challenges of MANETs, Ad Hoc Wireless Internet, MAC protocols for Ad hoc Wireless Networks-Issues, Design Goals and Classifications of the MAC Protocols.

UNIT II:

Routing Protocols for Ad Hoc Wireless Networks- Issues in Designing a Routing Protocol, Classifications of Routing Protocols, Topology-based versus Position-based Approaches, Issues and design goals of a Transport layer protocol, Classification of Transport layer solutions, TCP over Ad hoc Wireless Networks, Solutions for TCP over Ad Hoc Wireless Networks, Other Transport layer protocols.

UNIT III:

Security protocols for Ad hoc Wireless Networks- Security in Ad hoc Wireless Networks, Network Security Requirements, Issues and Challenges in Security Provisioning, Network Security Attacks, Key Management, Secure Routing in Ad hoc Wireless Networks, Cooperation in MANETs, Intrusion Detection Systems.

UNIT IV:

Basics of Wireless Sensors and Applications- The Mica Mote, Sensing and Communication Range, Design Issues, Energy Consumption, Clustering of Sensors, Applications, Data Retrieval in Sensor Networks-Classification of WSNs, MAC layer, Routing layer, Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs.

UNIT V:

Security in WSNs- Security in WSNs, Key Management in WSNs, Secure Data Aggregation in WSNs, Sensor Network Hardware-Components of Sensor Mote, Sensor Network Operating Systems–TinyOS, LA-TinyOS, SOS, RETOS, Imperative Language-nesC, **Dataflow Style Language-**TinyGALS, Node-Level Simulators, NS-2 and its sensor network extension, TOSSIM.

Text Books:

1. Ad Hoc Wireless Networks – Architectures and Protocols, 1st edition, C. Siva Ram Murthy, B. S. Murthy, Pearson Education, 2004

2. Ad Hoc and Sensor Networks – Theory and Applications, 2nd edition *Carlos Corderio Dharma P.Aggarwal,* World Scientific Publications / Cambridge University Press, March 2006

Reference Books:

1. Wireless Sensor Networks: An Information Processing Approach, 1st edition*, Feng Zhao, Leonidas Guibas*, Elsevier Science imprint, Morgan Kauffman Publishers, 2005, rp2009
2. Wireless Ad hoc Mobile Wireless Networks – Principles, Protocols and Applications, 1st edition*,* Subir Kumar Sarkar, et al., Auerbach Publications, Taylor & Francis Group, 2008
3. Ad hoc Networking, 1st edition*,Charles E.Perkins*, Pearson Education, 2001
4. Wireless Ad hoc Networking, 1st edition*, Shih-Lin Wu, Yu-Chee Tseng,* Auerbach Publications, Taylor & Francis Group, 2007
5. Wireless Sensor Networks – Principles and Practice, 1st edition, Fei Hu, Xiaojun Cao, An Auerbach book, CRC Press, Taylor & Francis Group, 2010

E-Resources:

1. https://archive.nptel.ac.in/courses/106/105/106105160/

III Year II Semester	2.Cyber Security (Professional Elective-II)	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the course, students will be able to:
CO1: Understand the basics of cyber security and different types of cyber-attacks.
CO2: Explain different mechanisms for password management.
CO3: Explain different techniques to prevent cyber-attacks.
CO4: Use of different techniques to secure internet browsing.
CO5: Understand security guidelines and standards for various mobile devices.

UNIT I:
Cyber Security Basics: Introduction, importance and objectives of cyber security. **Types of cyber-attacks:** Denial of Service (DoS), Distributed Denial of Service (DDoS) Man-in-the-Middle (MITM) Attacks, Crypto jacking, SQL Injection, Spamming Cyberterrorism Digital Property Misappropriation, Zero-Day Exploitation, Phishing, Digital Vandalism, Cyberstalking, Cyber Frauds and Forgery. **Types of computer malware:** Viruses, Trojan Horse, Rootkit, Spyware, Worms, Adware, Scareware, Browser Hijacker

UNIT II:
Securing your computer: Firewall Settings, Antivirus Software, Anti-Spyware Software, Anti-Spam Software , Security Updates, Secure Browsing Settings, Scan Devices before Data Transfer, Social Engineering Attack Precautions, Password management, Miscellaneous tips.
Password Management: Basics of Passwords, Effective Password Management Tips, Creating and Managing Secure Passwords, Strong Password, Use of Biometrics, Two-Factor, Authentication, Multi-Factor Authentication, One-Time Password (OTP) , Using Password Managers, Password Manager Tools.

UNIT III:
Prevention from cyber-attacks: Algorithms and Techniques, Firewalls, Intrusion Detection/ Prevention Systems, Authentication Using Hash, Multi-Factor Authentication, Mac Computer Firewall Configuration, Secure Sockets Layer , Virtual Private Network (Chapter 8)

UNIT IV:
Wireless Network Security: LAN Vulnerabilities, Wireless WAN Vulnerabilities, IoT vulnerabilities, wireless network security measures
Secure online shopping and Internet Browsing: HTTP Websites, Secure Electronic Transactions, Web Fraud Detection Systems, Browser Cache Clearing, Frequent Password Changes, Updating OS, Updating APPs

UNIT V:
Mobile Device Security : Importance of Mobile Security, types of Mobile Platforms ,Android Phone Security Guidelines, iPhone Security Guidelines, Windows Phone Security Guidelines, Mobile Application Management, Role based Access control, Open authorization, NIST guidelines
Cybersecurity Standards : ISO/IEC 27001 & 27002 Standards, Information Security Forum (ISF) Standards ,Payment Card Industry Data Security Standard (PCI/DSS)

Text Books:

1. Dr Kutub Thakur , Dr Al-Sakib Khan Pathan, "*Cybersecurity Fundamentals: A Real-World Perspective"*, CRC Press, Taylor & Francis Group,1st Edition ,2020.

Reference Books:

1. James Graham,Richard Howard,Ryan Olson, "*Cyber Security Essentials*", CRC Press,Taylor & Francis Group, 1st Edition, Reprint 2017.
2. Charles J. Brooks,Christopher Grow,Philip Craig,Donald Short, "*Cyber Security Essentials*", Sybex Publishers, October 2018.(ISBN: 9781119362395)
3. William Stallings and Lawrie Brown, "Computer Security Principles and Practice", Pearson, 3rd Edition, 2017

E-Resources:

1. https://in.coursera.org/speciali`zations/cyber-security
2. https://www.coursera.org/learn/intro-cyber-security-business
3. https://www.csoonline.com/article/3216404/what-is-oauth-how-the-open-authorization-framework-works.html

III Year II Semester	3.DevOps (Professional Elective-II)	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the course, students will be able to
CO1: Understand different actions performed through Version control tools like Git.
CO2: Perform Continuous Integration and Continuous Testing and Continuous Deployment using Jenkins by building and automating test cases using Maven & Gradle.
CO3: Ability to Perform Automated Continuous Deployment.
CO4: Ability to do configuration management using Ansible.
CO5: Understand to leverage Cloud-based DevOps tools using Azure DevOps.

Unit I: Introduction to DevOps:
DevOps Essentials - Introduction To AWS, GCP, Azure - Version control systems: Git and GitHub.

Unit II: Compile and Build Using Maven & Gradle
Introduction, Installation of Maven, POM files, Maven Build lifecycle, Build phases(compile build, test, package) Maven Profiles, Maven repositories(local, central, global),Maven plugins, Maven create and build Artificats, Dependency management, Installation of Gradle, Understand build using Gradle.

Unit III Continuous Integration using Jenkins
Install & Configure Jenkins, Jenkins Architecture Overview, Creating a Jenkins Job, Configuring a Jenkins job, Introduction to Plugins, Adding Plugins to Jenkins, Commonly used plugins (Git Plugin, Parameter Plugin, HTML Publisher, Copy Artifact and Extended choice parameters), Configuring Jenkins to work with java, Git and Maven, Creating a Jenkins Build and Jenkins workspace.

Unit IV: Configuration Management using Ansible
Ansible Introduction, Installation, Ansible master/slave configuration, YAML basics, Ansible modules, Ansible Inventory files, Ansible playbooks, Ansible Roles, adhoc commands in ansible.

Unit V: Building DevOps Pipelines using Azure
Create GitHub Account, Create Repository, Create Azure Organization, Create a new pipeline, Build a sample code, Modify azure-pipelines. yaml file.

Text Books:
1. Roberto Vormittag, "A Practical Guide to Git and GitHub for Windows Users: From Beginner to Expert in Easy Step-By-Step Exercises", Second Edition, Kindle Edition, 2016.
2. Jason Cannon, "Linux for Beginners: An Introduction to the Linux Operating System and Command Line", Kindle Edition, 2014.

Reference Books:
1. Hands-on Azure DevOps: CICD Implementation for Mobile, Hybrid, and Web Applications Using Azure DevOps and Microsoft Azure (English Edition) Paperback – 1 January 2020 by Mitesh Soni.
2. Jeff Geerling, "Ansible for DevOps: Server and configuration management for humans", First Edition, 2015.
3. David Johnson, "Ansible for DevOps: Everything You Need to Know to Use Ansible for DevOps", Second Edition, 2016.
4. Mariot Tsitoara, "Ansible 6. Beginning Git and GitHub: A Comprehensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019.

E-Resources:

1. https://www.jenkins.io/user-handbook.pdf
2. https://maven.apache.org/guides/getting-started/

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III Year II Semester	4.Cloud Computing (Professional Elective-II)	L	T	P	C
		3	0	0	3

Course Code: 23CS6E03

Course Outcomes:

At the end of the course, students will be able to:

CO1: Understand the fundamental concepts of cloud computing, including its architecture, service models, deployment models, characteristics, and major service providers.

CO2: Explain the foundational technologies enabling cloud computing such as parallel and distributed computing, SOA, web services, and virtualization.

CO3: Analyse the role of virtualization in cloud computing and differentiate between various cloud deployment models and architectures.

CO4: Evaluate the key challenges in cloud computing including scalability, interoperability, energy efficiency, and security across deployment models.

CO5: Apply knowledge of advanced cloud computing concepts to real-world applications such as healthcare, geoscience, and business, including the use of green and quantum cloud models.

UNIT I:

Introduction to Cloud Computing Fundamentals: Cloud Computing at a Glance, The Vision of a Cloud Computing, Defining A Cloud, Cloud Computing Reference Model, Types of Services (IAAS, PAAS, SAAS), Cloud Deployment Models (Public, Private, Hybrid), Service-Oriented Computing, Utility Computing, Cloud Computing Characteristics and Benefits, Cloud Service Providers (Amazon Web Services, Microsoft Azure, Google AppEngine).

UNIT II: Cloud Enabling Technologies

Eras of Computing, Ubiquitous Internet, Parallel and Distributed Computing, Elements of Parallel Computing, Hardware Architectures for Parallel Computing (SISD, SIMD, MISD, MIMD), Elements of Distributed Computing, Inter-Process Communication Technologies for Distributed Computing, Remote Procedure Calls (RPC), Service-Oriented Architecture (SOA), Web Services, Virtualization.

UNIT III: Virtualization and Containers

Characteristics Of Virtualized Environments, Taxonomy of Virtualization Techniques, Virtualization and Cloud Computing, Pros and Cons of Virtualization, Technology Examples (XEN, VMWARE), (Deleting the Content - Cloud Computing Architecture: Introduction, Cloud Reference Model Types of Clouds. Public Clouds: Private Clouds, Hybrid Clouds, Community Clouds

UNIT IV: Cloud computing challenges

Economics of the Cloud, Cloud Interoperability and Standards, Scalability and Fault Tolerance, Energy Efficiency in Clouds, Federated Clouds, Cloud Computing Security, Fundamentals of Computer Security, Cloud Security Architecture, Cloud Shared Responsibility Model, Security in Cloud Deployment Models.

UNIT V: Advanced Concepts in Cloud Computing

Quantum Cloud Computing. Energy Efficiency in Clouds, Energy-Efficient and Green Cloud Computing Architecture, Market Based Management of Clouds, Market-Oriented Cloud Computing, A Reference Model for MOCC, Cloud Applications: Healthcare: ECG Analysis in the Cloud, Geoscience: Satellite Image Processing, Business and Consumer Applications: CRM and ERP, Productivity, Social Networking

Text Books:

1. Mastering Cloud Computing, 2nd edition, Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi, Sivananda Poojara, Satish N. Srirama, Mc Graw Hill, 2024.

2. Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.

Reference Books:

1. Cloud Computing, Theory and Practice, Dan C Marinescu, 2nd edition, MK Elsevier, 2018.
2. Essentials of cloud Computing, K. Chandrasekharan, CRC press, 2014.
3. Online documentation and tutorials from cloud service providers (e.g., AWS, Azure, GCP)

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc21_cs14/preview.
2. https://www.coursera.org/specializations/cloud-computing
3. https://www.coursera.org/learn/introduction-to-cloud

III Year II Semester	1. Software Project Management (Professional Elective-III)	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the Course, students will be able to
CO1: Understand the evolution of software economics and compare conventional and modern software management principles with a focus on improving software development efficiency.
CO2: Identify and describe various phases of the software development life cycle and the associated artifacts used for effective project management and engineering.
CO3: Analyze model-based software architectures, workflows, and planning strategies to develop efficient and iterative project plans.
CO4: Apply project organization strategies, process automation techniques, and software metrics to manage and monitor software development projects.
CO5: Evaluate the implementation of Agile and DevOps practices, including Scrum, deployment pipelines, and DevOps tools, for continuous delivery and project agility.

UNIT I:
Conventional Software Management: The waterfall model, conventional software Management performance.
Evolution of Software Economics: Software Economics, pragmatic software cost estimation.
Improving Software Economics: Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.
The old way and the new: The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process.

UNIT II:
Life cycle phases: Engineering and production stages, inception, Elaboration, construction, transition phases.
Artifacts of the process: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

UNIT III:
Model based Software Architectures: A Management perspective and technical perspective.
Work Flows of the process: Software process workflows, Iteration workflows.
Checkpoints of the process: Major mile stones, Minor Milestones, Periodic status assessments.
Iterative Process Planning: Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

UNIT IV:
Project Organizations and Responsibilities: Line-of-Business Organizations, Project Organizations, evolution of Organizations.
Process Automation: Automation Building blocks, The Project Environment.
Project Control and Process instrumentation: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation.

UNIT V:
Agile Methodology, Adapting to Scrum, Patterns for Adopting Scrum, Iterating towards Agility.
Fundamentals of DevOps: Architecture, Deployments, Orchestration, Need, Instance of applications,

DevOps delivery pipeline, DevOps eco system. DevOps adoption in projects: Technology aspects, Agiling capabilities, Tool stack implementation, People aspect, processes

Text Books:

1. Software Project Management, Walker Royce, PEA, 2005.
2. Succeeding with Agile: Software Development Using Scrum, Mike Cohn, Addison Wesley.
3. The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations, Gene Kim, John Willis, Patrick Debois, Jez Humb,1st Edition, O'Reilly publications, 2016.

Reference Books:

1. Software Project Management, Bob Hughes,3/e, Mike Cotterell, TMH
2. Software Project Management, Joel Henry, PEA
3. Software Project Management in practice, Pankaj Jalote, PEA, 2005,
4. Effective Software Project Management, Robert K.Wysocki, Wiley,2006.
5. Project Management in IT, Kathy Schwalbe, Cengage

E-Resources:

1. https://nptel.ac.in/courses/106105218
2. https://www.coursera.org/learn/devops-culture-and-mindset
3. https://nptel.ac.in/courses/106106116

III Year II Semester	2.Natural Language Processing (Professional Elective-III)	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the course, students will be able to:
CO1: Understand the foundational concepts of Natural Language Processing, including language modeling techniques, regular expressions, and morphological analysis.
CO2: Apply statistical techniques like N-gram modeling, smoothing, and tagging methods for word-level text analysis.
CO3: Analyze syntactic structures of natural language using formal grammars and parsing algorithms to handle syntactic ambiguity.
CO4: Evaluate semantic representations and methods for word sense disambiguation using both supervised and unsupervised approaches.
CO5: Implement discourse analysis methods, including anaphora and coreference resolution, using established algorithms and lexical resources.

UNIT I: INTRODUCTION: Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.

UNIT II: WORD LEVEL ANALYSIS: Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part- of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.

UNIT III: SYNTACTIC ANALYSIS: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures

UNIT IV: SEMANTICS AND PRAGMATICS: Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.

UNIT V: DISCOURSE ANALYSIS AND LEXICAL RESOURCES: Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, other resources. Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).

Text Books:
1. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, 2nd Edition, Daniel Jurafsky, James H. Martin -Pearson Publication,2014.
2. Natural Language Processing with Python, First Edition, Steven Bird, Ewan Klein and Edward Loper, OReilly Media,2009.

Reference Books:
1. Language Processing with Java and Ling Pipe Cookbook, 1st Edition, Breck Baldwin, Atlantic Publisher, 2015.

2. Natural Language Processing with Java, 2nd Edition, Richard M Reese, OReilly Media,2015.
3. Handbook of Natural Language Processing, Second, Nitin Indurkhya and Fred J. Damerau, Chapman and Hall/CRC Press, 2010.Edition
4. Natural Language Processing and Information Retrieval, 3rd Edition, Tanveer Siddiqui, U.S. Tiwary, Oxford University Press,2008.

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc23_cs45/preview

III Year II Semester	3.Big Data Analytics (Professional Elective-III)	L	T	P	C
		3	0	0	3

Course Code: 23CS6E06

Course Outcomes:
At the end of the course, students will be able to
CO1: Understand the core concepts of Big Data and its applications across industries such as marketing, healthcare, finance, and trading.
CO2: Demonstrate knowledge of NoSQL databases, data models (key-value, document, graph), and work with distributed systems like Cassandra for handling large-scale data.
CO3: Apply data analysis techniques using Hadoop, including its architecture, file system (HDFS), and processing framework (MapReduce), and work with data query tools like Hive.
CO4: Analyze and process big data efficiently using Apache Spark, including RDDs, Data Frames, transformations, Spark SQL, and deployment models.
CO5: Implement real-time data processing using Spark Structured Streaming, understand event-time handling, watermarks, and transformations on live data streams.

UNIT I: Big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.

UNIT II: Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schema less databases, materialized views, distribution models, sharding, master-slave replication, peer- peer replication, shardingand replication, consistency, relaxing consistency, version stamps, working with Cassandra, Table creation, loading and reading data.

UNIT III: Data formats, analysing data with Hadoop, scaling out, Architecture of Hadoop distributed file system (HDFS), fault tolerance, with data replication, High availability, Data locality, Map Reduce Architecture, Process flow, Java interface, data flow, Hadoop I/O, data integrity, compression, serialization. Introduction to Hive, data types and file formats, HiveQL data definition, HiveQL data manipulation, Logical joins, Window functions, Optimization, Table partitioning, Bucketing, Indexing, Join strategies.

UNIT IV: Apache spark- Advantages over Hadoop, lazy evaluation, In memory processing, DAG, Spark context, Spark Session, RDD, Transformations- Narrow and Wide, Actions, Data frames ,RDD to Data frames, Catalyst optimizer, Data Frame Transformations, Working with Dates and Timestamps, Working with Nulls in Data, Working with Complex Types, Working with JSON, Grouping, Window Functions, Joins, Data Sources, Broadcast Variables, Accumulators, Deploying Spark- On-Premises Cluster Deployments, Cluster Managers- Standalone Mode, Spark on YARN , Spark Logs, The Spark UI- Spark UI History Server, Debugging and Spark First Aid

UNIT V: Spark-Performance Tuning, Stream Processing Fundamentals, Event-Time and State full Processing - Event Time, State full Processing, Windows on Event Time- Tumbling Windows, Handling Late Data with Watermarks, Dropping Duplicates in a Stream, Structured Streaming Basics - Core Concepts, Structured Streaming in Action, Transformations on Streams, Input and Output.

Text Books:

1. Big Data, Big Analytics: Emerging, Michael Minnelli, Michelle Chambers, and AmbigaDhiraj, 1st edition ,2013
2. SPARK: The Definitive Guide, Bill Chambers &MateiZaharia, O'Reilley, 2018-first Edition.
3. Business Intelligence and Analytic Trends for Today's Businesses", Wiley, First edition-2013.
4. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World Polyglot Persistence", Addison-Wesley Professional, 2012
5. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012

Reference Books:

1. "Hadoop Operations", O'Reilley, Eric Sammer, First Edition -2012.
2. "Programming Hive", O'Reilley, E. Capriolo, D. Wampler, and J. Rutherglen, 2012.
3. "HBase: The Definitive Guide", O'Reilley, Lars George, September 2011: First Edition..
4. "Cassandra: The Definitive Guide", O'Reilley, Eben Hewitt, 2010."Programming Pig", O'Reilley, Alan Gates, October 2011: First Edition

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc20_cs92/preview

III Year II Semester	4.Distributed Operating System (Professional Elective-III)	L	T	P	C
		3	0	0	3

Course Outcomes:
At the end of the course, students will be able to
CO1: Understand the fundamentals and architecture of distributed computing systems and analyze the key features and issues in message-passing communication.
CO2: Describe the working of Remote Procedure Call mechanisms and evaluate their implementation, transparency, and performance in distributed systems.
CO3: Explain the architecture and consistency models of Distributed Shared Memory systems and apply synchronization techniques to ensure correctness in distributed computing.
CO4: Analyze and compare resource management techniques including global scheduling, load balancing, and process migration in distributed systems.
CO5: Evaluate the design principles of distributed file systems and assess their reliability, performance, and fault tolerance mechanisms.

Unit I: Fundamentals: What is Distributed Computing Systems? Evolution of Distributed Computing System; Distributed Computing System Models; What is Distributed Operating System? Issues in Designing a Distributed Operating System; Introduction to Distributed Computing Environment (DCE).
Message Passing: Introduction, Desirable features of a Good Message Passing System, Issues in PC by Message Passing, Synchronization, Buffering, Multi-datagram Messages, Encoding and Decoding of Message Data, Process Addressing, Failure Handling, Group Communication, Case Study: 4.3 BSD UNIX IPC Mechanism.

Unit II: Remote Procedure Calls: Introduction, The RPC Model, Transparency of RPC, Implementing RPC Mechanism, Stub Generation, RPC Messages, Marshaling Arguments and Results, Server Management, Parameter-Passing Semantics, Call Semantics, Communication Protocols for RPCs, Complicated RPCs, Client-Server Binding, Exception Handling, Security, Some Special Types of RPCs, RPC in Heterogeneous Environments, Lightweight RPC, Optimization for Better Performance, Case Studies: Sun RPC

Unit III: Distributed Shared Memory: Introduction, General Architecture of DSM systems, Design and Implementation Issues of DSM, Granularity, Structure of Shared Memory Space, Consistency Models, Replacement Strategy, Thrashing, Other approaches to DSM, Heterogeneous DSM, Advantages of DSM. Synchronization: Introduction, Clock Synchronization, Event Ordering, Mutual Exclusion, Dead Lock, Election Algorithms

Unit IV: Resource Management: Introduction, Desirable Features of a Good Global Scheduling Algorithm, Task Assignment Approach, Load – Balancing Approach, Load – Sharing Approach Process Management: Introduction, Process Migration, Threads.

Unit V: Distributed File Systems: Introduction, Desirable Features of a Good Distributed File System, File models, File–Accessing Models, File – Sharing Semantics, File – Caching Schemes, File Replication, Fault Tolerance, Atomic Transactions and Design Principles.

Text books
1. Pradeep. K. Sinha: Distributed Operating Systems: Concepts and Design, PHI, 2007.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, Pearson Education, 2013.
2. Ajay D. Kshemkalyani and MukeshSinghal, Distributed Computing: Principles, Algorithms and Systems, Cambridge University Press, 2008
3. Sunita Mahajan, Seema Shan, " Distributed Computing", Oxford University Press,2015

E-Resources:

1. https://onlinecourses.nptel.ac.in/noc21_cs87/preview

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III Year II Semester	Cryptography & Network Security Lab	L	T	P	C
		0	0	3	1.5

Course Code: 23CS6L01

Course Outcomes:
At the end of the course, students will be able to
CO1: Apply basic bitwise operations (XOR, AND) in C to manipulate strings and understand low-level cryptographic transformations.
CO2: Implement classical encryption techniques like Caesar Cipher, Substitution Cipher, and Hill Cipher in Java to understand the foundations of cryptography.
CO3: Develop programs in Java/C to simulate modern symmetric encryption algorithms such as DES, Blowfish, and Rijndael (AES).
CO4: Utilize Java Cryptography Architecture (JCA) to implement encryption and key management, demonstrating the use of tools like Java KeyTool.
CO5: Implement public key cryptography algorithms such as RSA and simulate the Diffie-Hellman Key Exchange using JavaScript and HTML to model secure communication.
CO6: Compute and verify message digests using cryptographic hash functions like SHA-1 to ensure data integrity in Java applications.

List of Experiments:

1. Write a C program that contains a string (char pointer) with a value \Hello World'. The program should XOR each character in this string with 0 and displays the result.
2. Write a C program that contains a string (char pointer) with a value \Hello World'. The program should AND or and XOR each character in this string with 127 and display the result
3. Write a Java program to perform encryption and decryption using the following algorithms:
    a) Ceaser Cipher
    b) Substitution Cipher
    c) Hill Cipher
4. Write a Java program to implement the DES algorithm logic
5. Write a C/JAVA program to implement the Blowfish algorithm logic
6. Write a C/JAVA program to implement the Rijndael algorithm logic.
7. Using Java Cryptography, encrypt the text "Hello world" using Blowfish. Create your own key using Java key tool.
8. Write a Java program to implement RSA Algorithm
9. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript. Consider the end user as one of the parties (Alice) and the JavaScript application as other party (bob).
10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.

Text books:
1. Stallings, W. (2017). *Cryptography and network security: Principles and practice* (7th ed.). Pearson.
2. Forouzan, B. A., & Mukhopadhyay, D. (2015). *Cryptography and network security* (2nd ed.). McGraw-Hill Education.

Reference Books:
1. Hook, D. (2005). *Beginning cryptography with Java*. Wiley Publishing, Inc.
2. Menezes, A. J., van Oorschot, P. C., & Vanstone, S. A. (1996). *Handbook of applied cryptography*. CRC Press.

E resources:

1. https://www.wiley.com/en-in/Beginning+Cryptography+with+Java-p-9780764596339

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III Year II Semester	Machine Learning Lab	L	T	P	C
		0	0	3	1.5

Course Code: 23IT6L02

Course Outcomes:

At the end of the course, students will be able to:

CO1: Understand the implementation procedures for the machine learning algorithms.
CO2: Design Java/Python programs for various Learning algorithms.
CO3: Apply appropriate data sets to the Machine Learning algorithms.
CO4: Identify and apply Machine Learning algorithms to solve real world problems.
CO5: Apply different Classification algorithms to solve real world problems.

Week-1:

Implementation of Python Basic Libraries such as Statistics, Math, NumPy and SciPy

a) Usage of methods such as floor(), ceil(), sqrt(), isqrt(), gcd() etc.
b) Usage of attributes of array such as ndim, shape, size, methods such as sum(), mean(), sort(), sin() etc.
c) Usage of methods such as det(), eig() etc.
d) Consider a list datatype(1D) then reshape it into 2D,3D matrix using NumPy
e) Generate random matrices using NumPy
f) Find the determinant of a matrix using SciPy
g) Find eigen value and eigen vector of a matrix using SciPy

Week2:

Implementation of Python Libraries for ML application such as Pandas and Matplotlib.

a) Create a Series using pandas and display
b) Access the index and the values of our Series
c) Compare an array using NumPy with a series using pandas
d) Define Series objects with individual indices
e) Accesssing le value of a series
f) Load datasets in a Data frame variable using pandas
g) Usage of different methods in Matplotlib.

Week3:

a) Creation and Loading different types of datasets in Python using the required libraries.
   i. Creation using pandas
   ii. Loading CSVdataset files using Pandas
   iii. Loading datasets using sklearn
b) Write a python program to compute Mean, Median, Mode, Variance, Standard Deviation using Datasets

Week4:

Write a Python program to implement Simple Linear Regression and plot the graph.

Week5:

Write a Python program to implement Logistic Regression for iris using sklearn and plot confusion matrix

Week6:
Implement Support Vector Machine for a dataset and compare the accuracy by applying The following kernel functions: i. Linear ii. Polynomial iii. RBF

Week7:
Write a python program to implement K-Means clustering Algorithm. Vary the number of k values as follows and compare the results: i. 1 ii. 3 iii. 5

Week8:
Build an Artificial Neural Network by implementing the Back-propagation algorithm and test the same using appropriate data sets

Week9:
Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a.CSV file. Compute the accuracy of the classifier, considering few test data sets.

Week10.
Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.

Text books:
1. Raschka, S., & Mirjalili, V. (2019). *Python machine learning* (3rd ed.). Packt Publishing.
2. Géron, A. (2019). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow* (2nd ed.). O'Reilly Media.

Reference books:
1. McKinney, W. (2018). *Python for data analysis: Data wrangling with Pandas, NumPy, and IPython* (2nd ed.). O'Reilly Media.
2. VanderPlas, J. (2016). *Python data science handbook: Essential tools for working with data*. O'Reilly Media.

E Resources:
- https://numpy.org/doc
- https://docs.scipy.org/doc/scipy
- https://www.tensorflow.org

III Year II Semester	Soft skills (Skill Enhancement Course)	L	T	P	C
		0	1	2	2

Course Code: 23SH6S01

Course Outcomes:
At the end of the course, students will be able to:
CO1: Develop analytical thinking, active listening, and effective verbal and non-verbal communication skills to build a confident self-image.
CO2: Demonstrate self-management strategies and professional etiquette to enhance personal and workplace effectiveness.
CO3: Apply basic grammar and writing techniques to communicate ideas clearly and professionally through various written formats.
CO4: Prepare for and participate effectively in job selection processes including group discussions, resume writing, and interviews.
CO5: Understand and apply strategies to build and maintain positive interpersonal relationships in personal and professional settings.

UNIT – I Analytical Thinking & Listening Skills: Self-Introduction, Shaping Young Minds - A Talk by Azim Premji (Listening Activity), Self – Analysis, Developing Positive Attitude, Perception.
Communication Skills: Verbal Communication; Non Verbal Communication (Body Language)

UNIT – II Self-Management Skills: Anger Management, Stress Management, Time Management, Six Thinking Hats, Team Building, Leadership Qualities
Etiquette: Social Etiquette, Business Etiquette, Telephone Etiquette, Dining Etiquette

UNIT – III Standard Operation Methods: Basic Grammars, Tenses, Prepositions, Pronunciation, Letter Writing; Note Making, Note Taking, Minutes Preparation, Email & Letter Writing

UNIT-IV Job-Oriented Skills: Group Discussion, Mock Group Discussions, Resume Preparation, Interview Skills, Mock Interviews

UNIT-V Interpersonal relationships: Introduction, Importance, Types, Uses, Factors affecting interpersonal relationships, Accommodating different styles, Consequences of interpersonal relationships

Text Books:
1. Barun K. Mitra, Personality Development and Soft Skills, Oxford University Press, 2011.
2. S.P. Dhanavel, English and Soft Skills, Orient Blackswan, 2010.

Reference Books:
1. R.S.Aggarwal, A Modern Approach to Verbal & Non-Verbal Reasoning, S.Chand& Company Ltd., 2018.
2. Raman, Meenakshi& Sharma, Sangeeta, Technical Communication Principles and Practice, Oxford University Press, 2011.

E-Resources:
https://swayam-plus.swayam2.ac.in/courses/course-details?id=P_CAMBR_01

KKR & KSR INSTITUUTE OF TECHNOLOGY AND SCIENCES (AUTONOMOUS)

(Accredited by NBA & NAAC with Grade "A" and Affiliated to JNTUK-Kakinada)
(Vinjanampadu, Vatticherukuru Mandal, Guntur-522017.Andhra Pradesh. INDIA)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

III Year II Semester	Technical Paper Writing & IPR	L	T	P	C
		2	0	0	-

Course Code: 23GE6M01

Course Outcomes:

At the end of the course, student will be able to:

CO1: Understand the principles of technical writing and apply appropriate language, structure, and formatting to plan and draft professional technical reports.

CO2: Draft technical documents using illustrations and revise them through structured editing, focusing on clarity, grammar, and professional language.

CO3: Demonstrate effective proofreading and summarization techniques and present final technical documents both in written and verbal forms.

CO4: Use advanced word processing tools for formatting, editing, referencing, and securing technical documents efficiently.

CO5: Understand the basics of Intellectual Property Rights and the process of innovation, patenting, and international cooperation in IP protection.

Unit I: Introduction: An introduction to writing technical reports, technical sentences formation, using transitions to join sentences, Using tenses for technical writing.
Planning and Structuring: Planning the report, identifying reader(s), Voice, Formatting and structuring the report, Sections of a technical report, Minutes of meeting writing.

Unit II: Drafting report and design issues: The use of drafts, Illustrations and graphics.
Final edits: Grammar, spelling, readability and writing in plain English: Writing in plain English, Jargon and final layout issues, Spelling, punctuation and Grammar, Padding, Paragraphs, Ambiguity.

Unit III: Proofreading and summaries: Proofreading, summaries, Activities on summaries. **Presenting final reports:** Printed presentation, Verbal presentation skills, Introduction to proposals and practice.

Unit IV: Using word processor: Adding a Table of Contents, Updating the Table of Contents, Deleting the Table of Contents, Adding an Index, Creating an Outline, Adding Comments, Tracking Changes, Viewing Changes, Additions, and Comments, Accepting and Rejecting Changes , Working with Footnotes and Endnotes, Inserting citations and Bibliography, Comparing Documents, Combining Documents, Mark documents final and make them read only., Password protect Microsoft Word documents., Using Macros,

Unit V: **Nature of Intellectual Property:** Patents, Designs, Trade and Copyright.
Process of Patenting and Development: technological research, innovation, patenting, development.
International Scenario: International cooperation on Intellectual Property

Text Books:

1. Kompal Bansal & Parshit Bansal, "Fundamentals of IPR for Beginner's", 1st Ed., BS Publications, 2016.
2. William S. Pfeiffer and Kaye A. Adkins, "Technical Communication: A Practical Approach", Pearson.
3. Ramappa,T., "Intellectual Property Rights Under WTO", 2nd Ed., S Chand, 2015.

Reference Books:

1. Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011.

2. Day R, How to Write and Publish a Scientific Paper, Cambridge University Press(2006)

E-Resources:

1. https://www.udemy.com/course/reportwriting/
2. https://www.udemy.com/course/professional-business-english-and-technical-report-writing/